Data Analysis on online Food Delivery

In [1]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

Data={  'Order_Id':[ 1001,1002,1003,1004,1005,1006,1007,1008],
        'Cust_Name':["Anuj Agrawal", "Vashu Banjaare", "Aryan Katakwar", "Akriti Soni", "Ayush Gupta", "Ritesh Danser
        'Order_Date':["8-08-2025", "9-08-2025", "8-08-2025", "11-08-2025", "8-08-2025", "9-08-2025", "11-08-2025", "9
        'Delivery_Time(min)':[25,30,20,26,30,35,25,45],
        'Food_Items':["Pizza","Pizza", "Sushi", "Biryani", "Fried Rice", "Sandwich","Sushi","Biryani"],
        'Restaurant_Name':["PizzaZone", "PizzaZone", "SushiWorld", "SpiceLand", "Asian Express", "FreshhEats", "Sushi
        'Delivery_Rating':[4.2,2.4,4.6,3.6,4.7,2.5,3.5, 3.9],
        'Food_Rating':[3.3, 2.7, 4.8, 2.5, 4.9, 3.1, 4.4, 2.3],
        'Feedback':["Average", "Below Average", "Perfect", "Worst", "Nice", "Not Bad", "Perfect", "Below Average"],
        'Price':[499, 499, 70, 100, 59, 69, 70, 100],
        'Payment_Mode':["Upi", "Credit Card", "Upi", "Debit Card", "Upi" ,"Credit Card", "Cash","Upi"],
        'Location':["New York", "New York", "Chicago", "Boston", "Seattle","san Francisco","Chicago", "Boston"]}

df=pd.DataFrame(Data)
print(df)
df.to_csv('FoodRecords2.csv', index=False)
```

```
   Order_Id        Cust_Name  Order_Date  Delivery_Time(min)  Food_Items  \
0      1001     Anuj Agrawal   8-08-2025                  25       Pizza   
1      1002   Vashu Banjaare   9-08-2025                  30       Pizza   
2      1003   Aryan Katakwar   8-08-2025                  20       Sushi   
3      1004     Akriti Soni  11-08-2025                  26     Biryani   
4      1005      Ayush Gupta   8-08-2025                  30  Fried Rice   
5      1006   Ritesh Dansena   9-08-2025                  35    Sandwich   
6      1007       Arya Gupta  11-08-2025                  25       Sushi   
7      1008  Saniya Banjaare   9-08-2025                  45     Biryani   

  Restaurant_Name  Delivery_Rating  Food_Rating       Feedback  Price  \
0       PizzaZone              4.2          3.3        Average    499   
1       PizzaZone              2.4          2.7  Below Average    499   
2      SushiWorld              4.6          4.8        Perfect     70   
3       SpiceLand              3.6          2.5          Worst    100   
4   Asian Express              4.7          4.9           Nice     59   
5      FreshhEats              2.5          3.1        Not Bad     69   
6      SushiWorld              3.5          4.4        Perfect     70   
7       SpiceLand              3.9          2.3  Below Average    100   

  Payment_Mode       Location  
0          Upi       New York  
1  Credit Card       New York  
2          Upi        Chicago  
3   Debit Card         Boston  
4          Upi        Seattle  
5  Credit Card  san Francisco  
6         Cash        Chicago  
7          Upi         Boston  
```

In [114…

```
# Data Cleaning
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8 entries, 0 to 7
Data columns (total 12 columns):
 #   Column              Non-Null Count  Dtype  
---  ------              --------------  -----  
 0   Order_Id            8 non-null      int64  
 1   Cust_Name           8 non-null      object 
 2   Order_Date          8 non-null      object 
 3   Delivery_Time(min)  8 non-null      int64  
 4   Food_Items          8 non-null      object 
 5   Restaurant_Name     8 non-null      object 
 6   Delivery_Rating     8 non-null      float64
 7   Food_Rating         8 non-null      float64
 8   Feedback            8 non-null      object 
 9   Price               8 non-null      int64  
 10  Payment_Mode        8 non-null      object 
 11  Location            8 non-null      object 
dtypes: float64(2), int64(3), object(7)
memory usage: 900.0+ bytes
```

In [115…

```
pd.isnull(df)
```

Out[115…

	Order_Id	Cust_Name	Order_Date	Delivery_Time(min)	Food_Items	Restaurant_Name	Delivery_Rating	Food_Rating	Feedbac
0	False	False	False	False	False	False	False	False	Fals
1	False	False	False	False	False	False	False	False	Fals
2	False	False	False	False	False	False	False	False	Fals
3	False	False	False	False	False	False	False	False	Fals
4	False	False	False	False	False	False	False	False	Fals
5	False	False	False	False	False	False	False	False	Fals
6	False	False	False	False	False	False	False	False	Fals
7	False	False	False	False	False	False	False	False	Fals

In [116…

```
pd.isnull(df).sum()
```

Out[116…

```
Order_Id             0
Cust_Name            0
Order_Date           0
Delivery_Time(min)   0
Food_Items           0
Restaurant_Name      0
Delivery_Rating      0
Food_Rating          0
Feedback             0
Price                0
Payment_Mode         0
Location             0
dtype: int64
```

In [117…

```
# Description od DataFrame
df.describe()
```

Out[117…

	Order_Id	Delivery_Time(min)	Delivery_Rating	Food_Rating	Price
count	8.00000	8.000000	8.000000	8.000000	8.00000
mean	1004.50000	29.500000	3.675000	3.500000	183.25000
std	2.44949	7.690439	0.868085	1.051529	195.44802
min	1001.00000	20.000000	2.400000	2.300000	59.00000
25%	1002.75000	25.000000	3.250000	2.650000	69.75000
50%	1004.50000	28.000000	3.750000	3.200000	85.00000
75%	1006.25000	31.250000	4.300000	4.500000	199.75000
max	1008.00000	45.000000	4.700000	4.900000	499.00000

In [118…

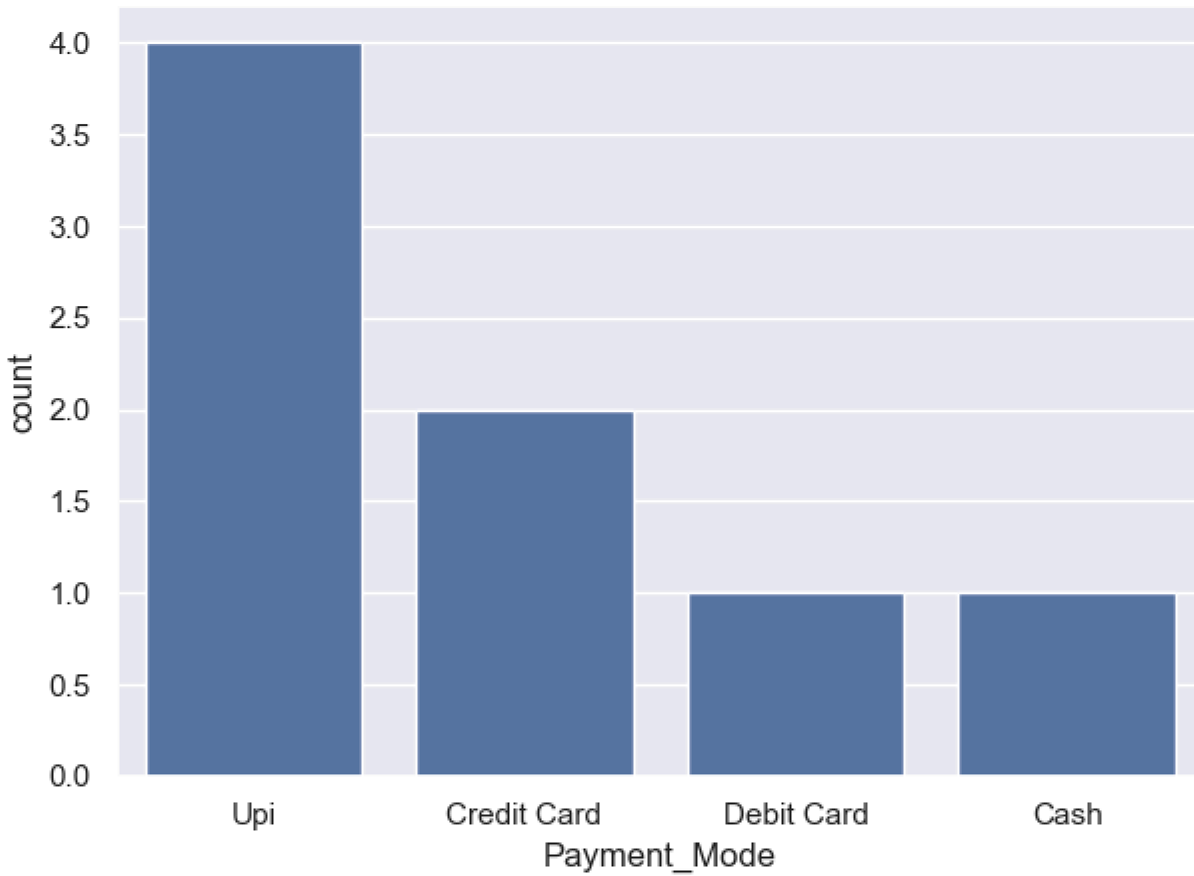
```
df[['Delivery_Rating','Food_Rating', 'Price']].describe()
```

Out[118…

	Delivery_Rating	Food_Rating	Price
count	8.000000	8.000000	8.00000
mean	3.675000	3.500000	183.25000
std	0.868085	1.051529	195.44802
min	2.400000	2.300000	59.00000
25%	3.250000	2.650000	69.75000
50%	3.750000	3.200000	85.00000
75%	4.300000	4.500000	199.75000
max	4.700000	4.900000	499.00000

In [119…

```
# Exploratory Data Analysis
ax=sns.countplot(x='Payment_Mode', data=df)
```



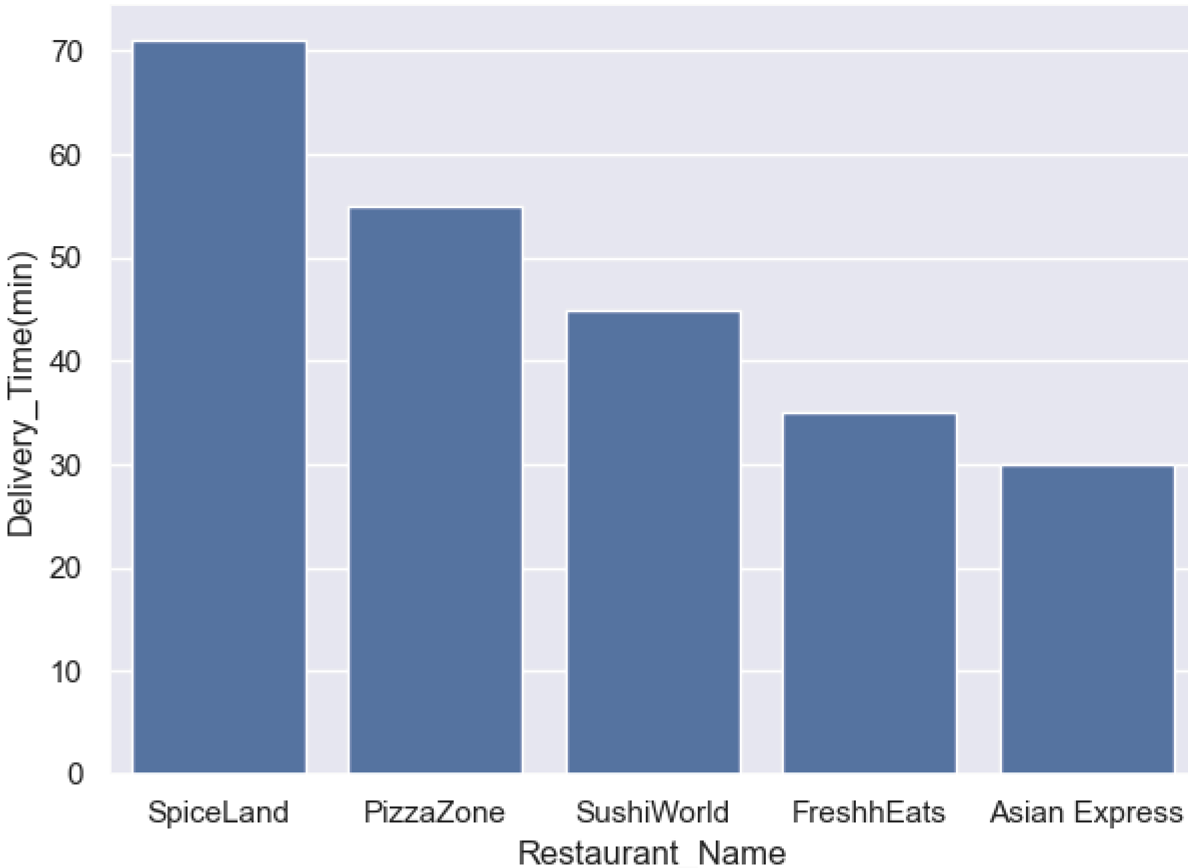
In [120…

```
datas=df.groupby(['Payment_Mode'], as_index=False)['Price'].sum().sort_values(by='Price', ascending=False)
datas
```

Out[120…

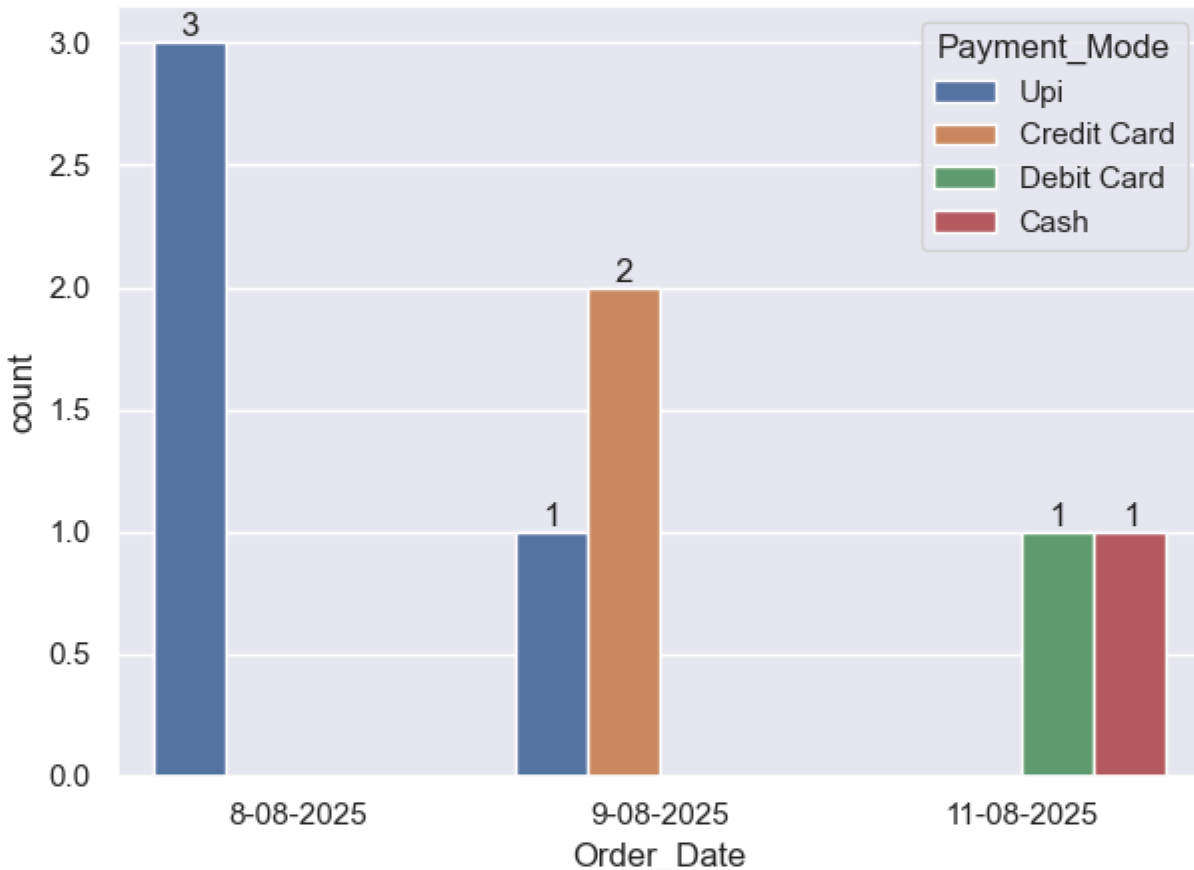
	Payment_Mode	Price
3	Upi	728
1	Credit Card	568
2	Debit Card	100
0	Cash	70

In [10]:
```
import seaborn as sns
sns.set(rc={'figure.figsize':(7,5)})
datas= df.groupby(['Restaurant_Name'],as_index=False)['Delivery_Time(min)'].sum().sort_values(by='Delivery_Time(min)
sns.barplot(x='Restaurant_Name', y= 'Delivery_Time(min)', data=datas)
plt.show()
```



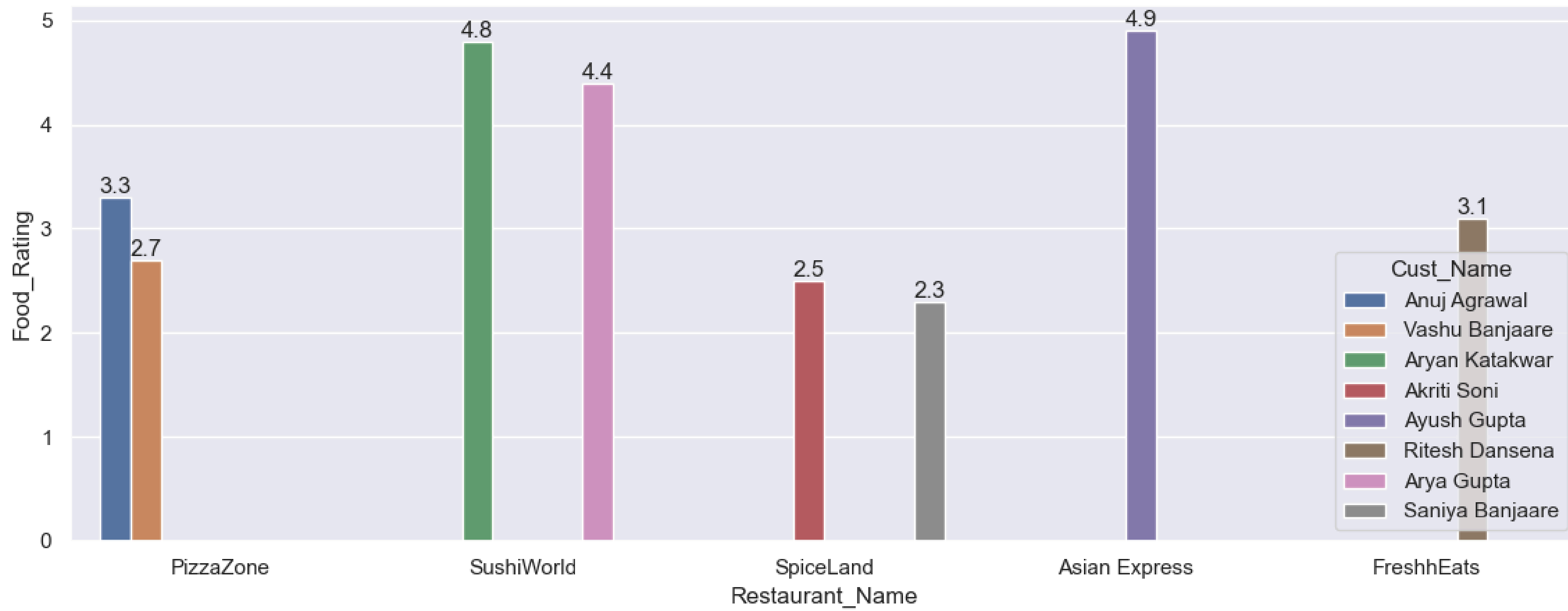
Analysis Of More Attributes

In [111…
```
ax=sns.countplot(x='Order_Date',data=df, hue='Payment_Mode')
for bars in ax.containers:
    ax.bar_label(bars)
```



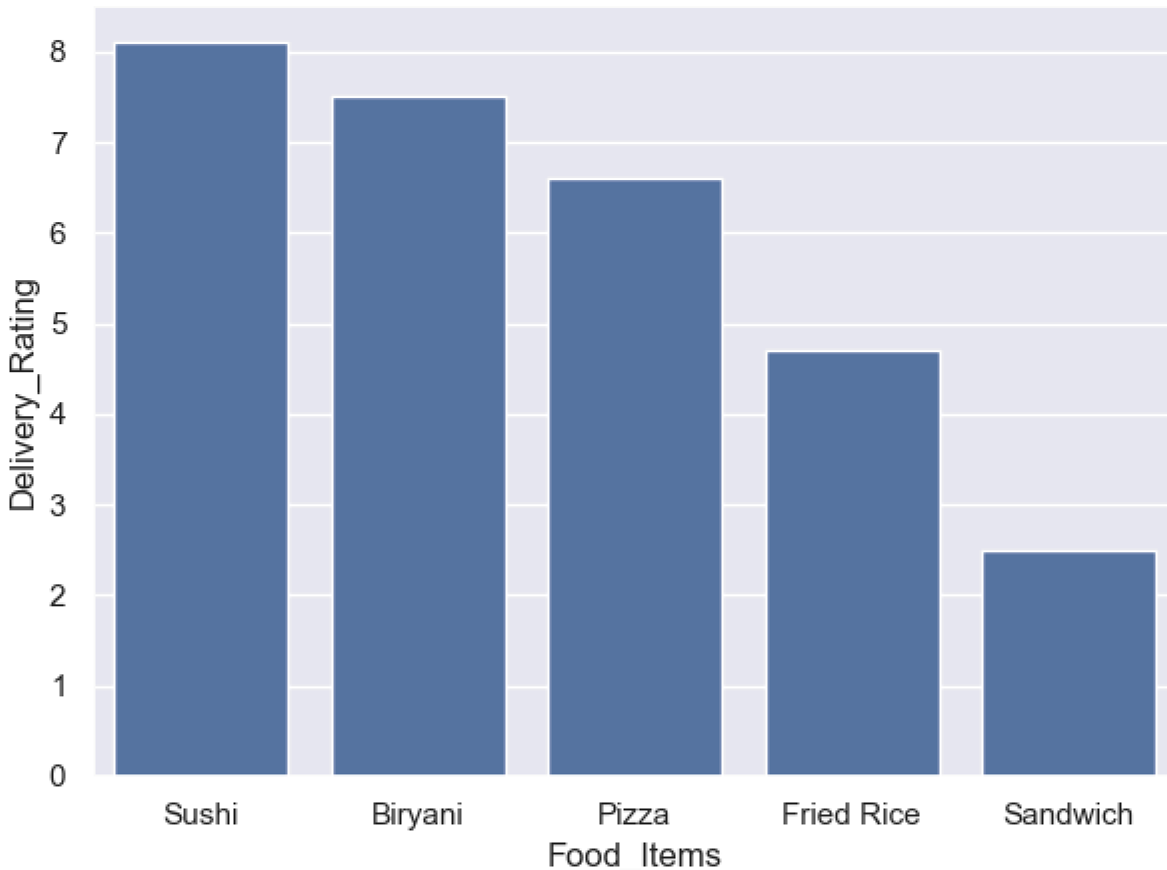
In [11]:

```
# Restaurant_Name with its food Rating
ax=sns.barplot(x='Restaurant_Name',y='Food_Rating', hue='Cust_Name',data=df)
for bars in ax.containers:
    ax.bar_label(bars)
```



In [13]:

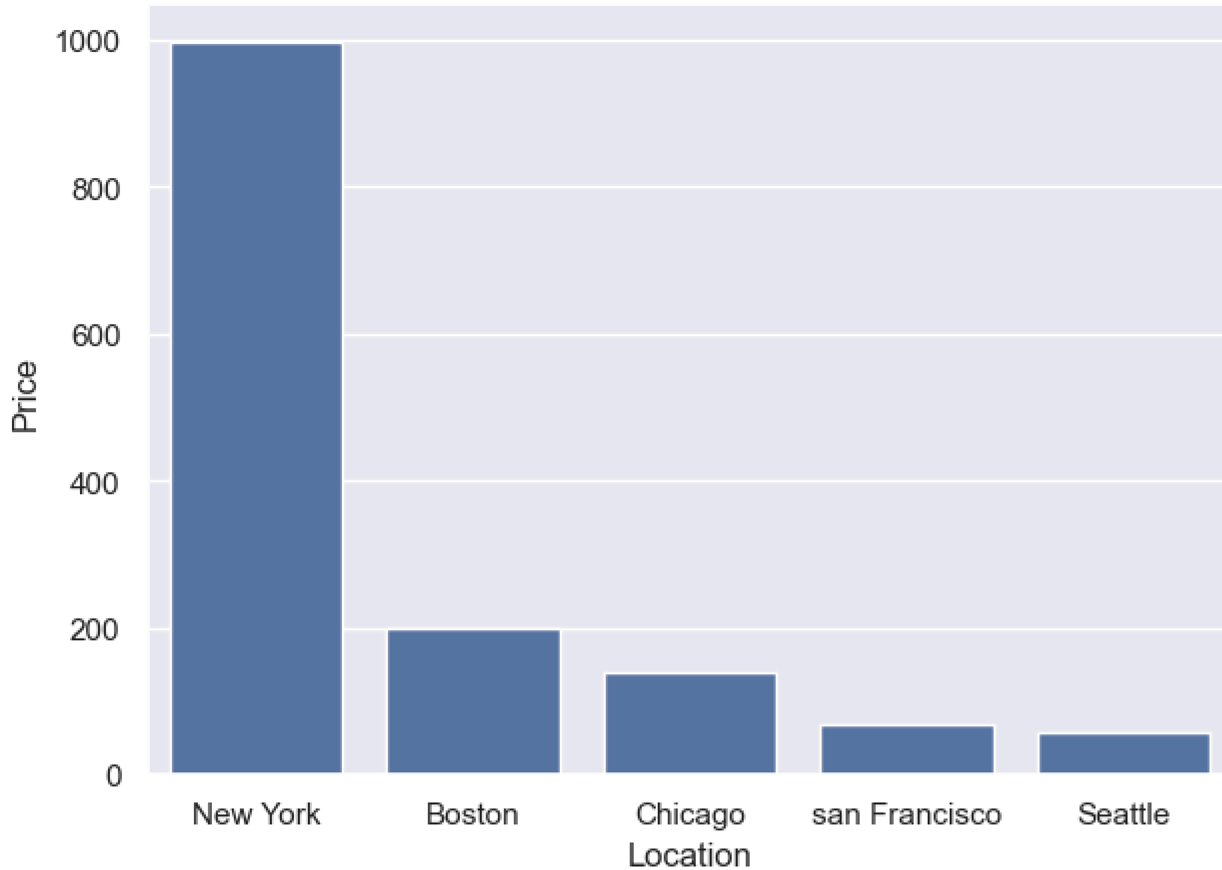
```
# Delivery_Rating vs Food_Items
import seaborn as sns
sns.set(rc={'figure.figsize':(7,5)})
Sales_Food= df.groupby(['Food_Items'],as_index=False)['Delivery_Rating'].sum().sort_values(by='Delivery_Rating',ascen
sns.barplot(x='Food_Items', y= 'Delivery_Rating', data=Sales_Food)
plt.show()
```



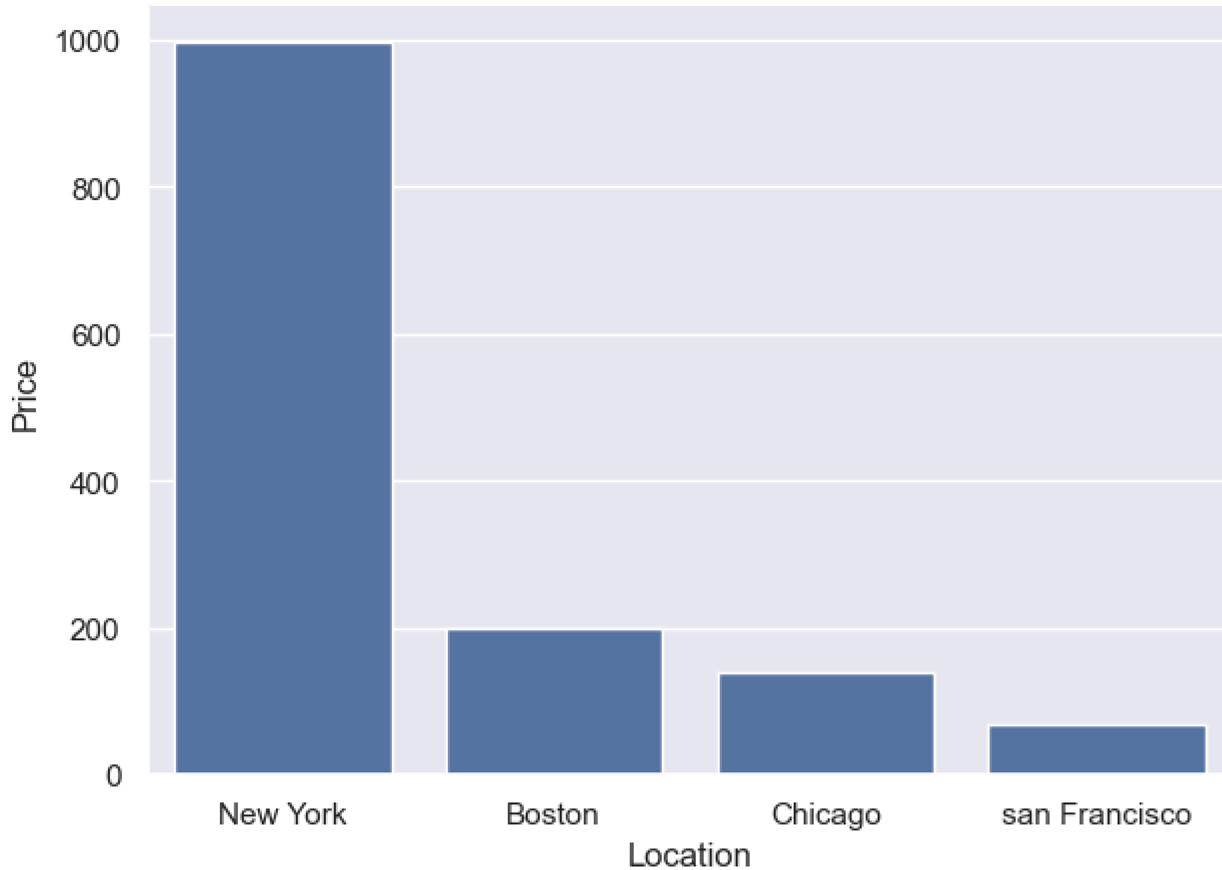
Analysis On City:

In [109…

```
import seaborn as sns
#  City Location vs Price
sns.set(rc={'figure.figsize':(7,5)})
Sales_Location= df.groupby(['Location'],as_index=False)['Price'].sum().sort_values(by='Price',ascending=False).head(5
sns.barplot(x='Location', y= 'Price', data= Sales_Location)
plt.show()
```

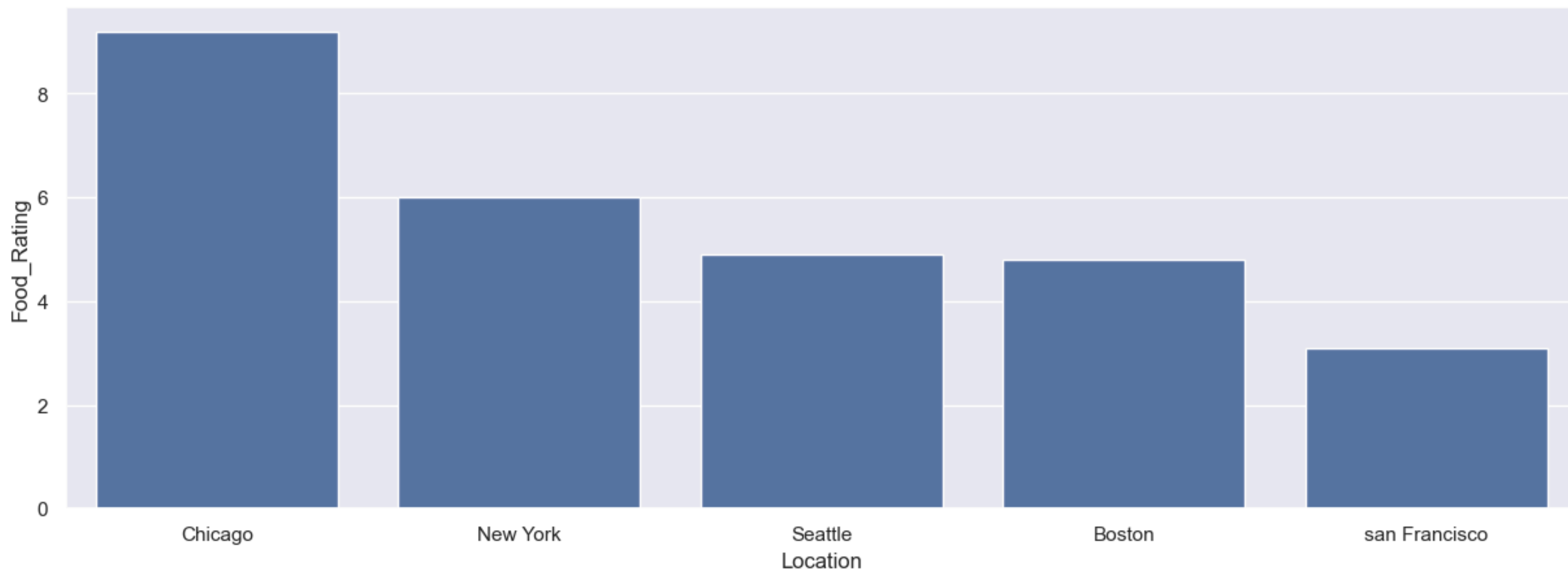


In [15]:
```
import seaborn as sns
#  City Location vs Price from 4 location
sns.set(rc={'figure.figsize':(7,5)})
Sales_Location= df.groupby(['Location'],as_index=False)['Price'].sum().sort_values(by='Price',ascending=False).head(4
sns.barplot(x='Location', y= 'Price', data= Sales_Location )
plt.show()
```



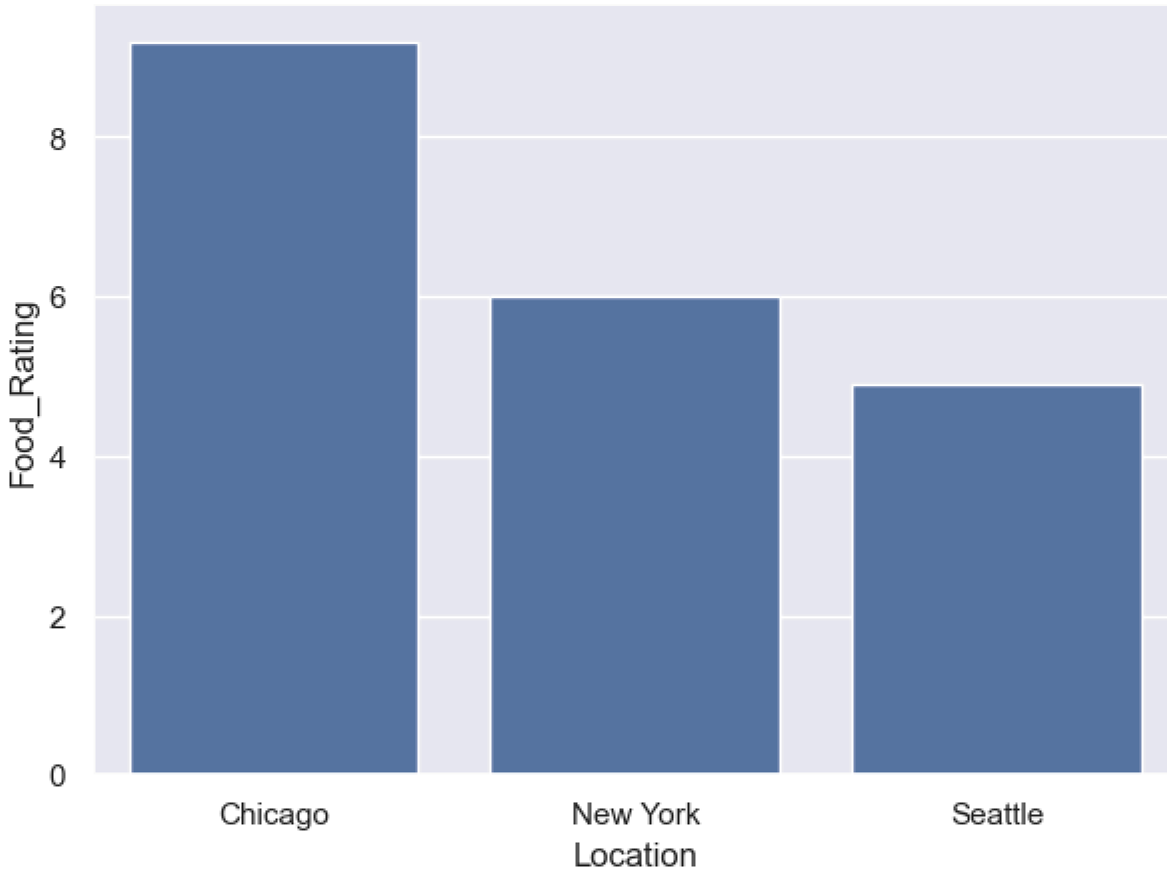
In [126…

```
import seaborn as sns
#  City Location With Food_Rating
sns.set(rc={'figure.figsize':(15,5)})
Sales_Location= df.groupby(['Location'],as_index=False)['Food_Rating'].sum().sort_values(by='Food_Rating',ascending=F
sns.barplot(x='Location', y= 'Food_Rating', data= Sales_Location )
plt.show()
```



In [127…

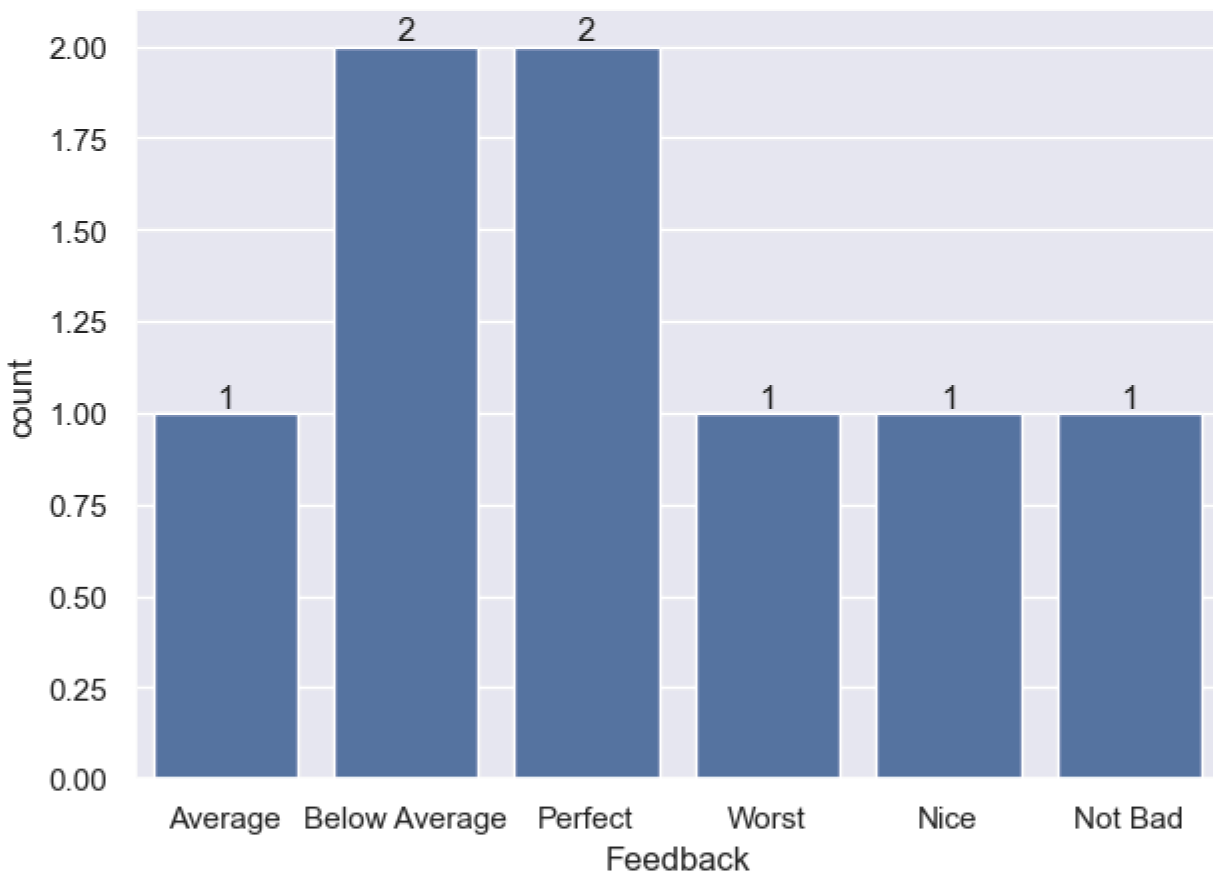
```
import seaborn as sns
#  City Location With Food_Rating from 3 locations
sns.set(rc={'figure.figsize':(7,5)})
Sales_Location= df.groupby(['Location'],as_index=False)['Food_Rating'].sum().sort_values(by='Food_Rating',ascending=F
sns.barplot(x='Location', y= 'Food_Rating', data= Sales_Location )
plt.show()
```



Analysis On Feedback:

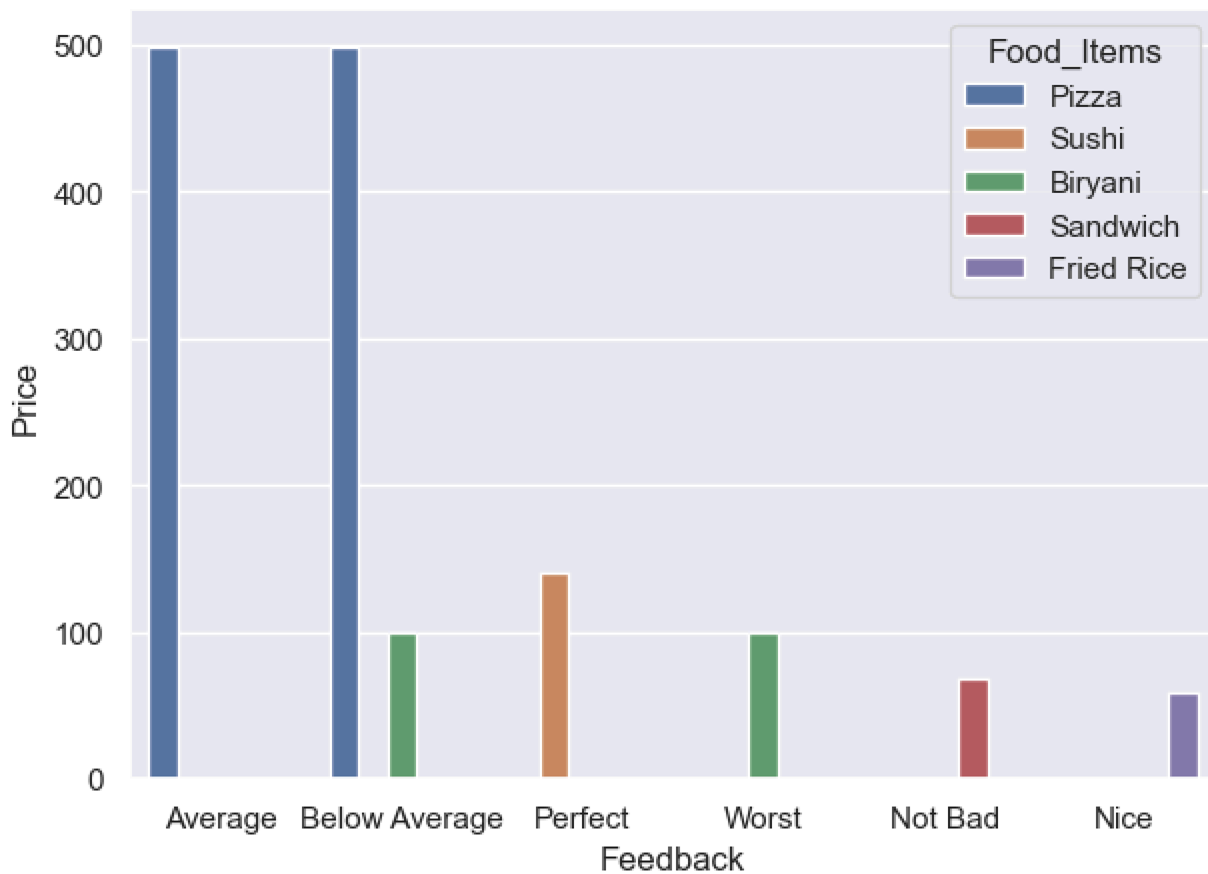
In [123…

```
import seaborn as sns
sns.set(rc={'figure.figsize':(7,5)})
ax=sns.countplot(x='Feedback', data=df)
for bars in ax.containers:
    ax.bar_label(bars)
```



In [124…

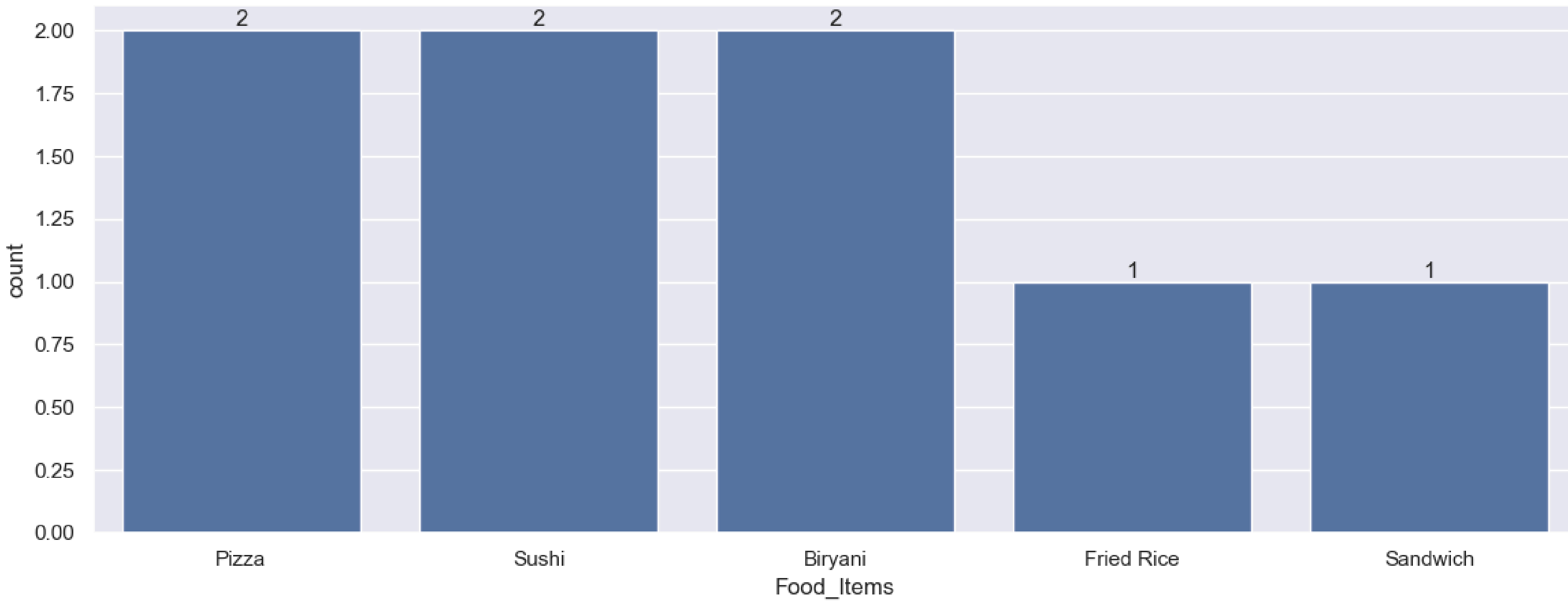
```
Sales_ms= df.groupby(['Feedback','Food_Items'],as_index=False)['Price'].sum().sort_values(by='Price',ascending=False)
sns.barplot(x='Feedback', y= 'Price', data= Sales_ms, hue='Food_Items')
plt.show()
```



Analysis Of Food_Items And Cust_Name :

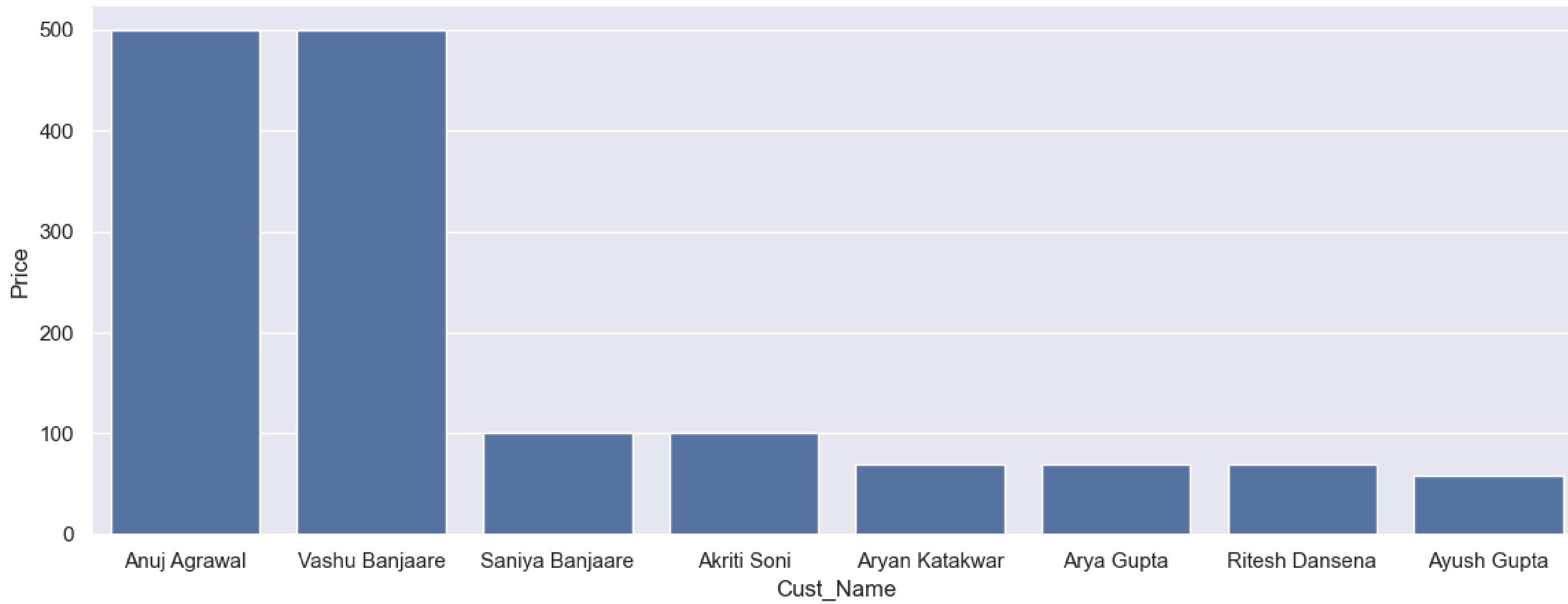
In [74]:

```
sns.set(rc={'figure.figsize':(14,5)})
ax=sns.countplot(x='Food_Items',data=df)
for bars in ax.containers:
    ax.bar_label(bars)
```



In [2]:

```
import seaborn as sns
#  City Location With Delivery_Rating from 3 Locations
sns.set(rc={'figure.figsize':(14,5)})
Sales_pc= df.groupby(['Cust_Name'],as_index=False)['Price'].sum().sort_values(by='Price',ascending=False)
sns.barplot(x='Cust_Name', y= 'Price', data= Sales_pc)
plt.show()
```



In [6]:

```
import seaborn as sns
import pandas as pd
df=pd.read_csv('FoodRecords2.csv')
sns.set(rc={'figure.figsize':(14,5)})
ax = sns.stripplot(x='Payment_Mode', y='Cust_Name', hue='Cust_Name',data=df)
for bars in ax.containers:
    ax.bar_label(bars)
```

In [1]:
```
print("End of Practice Dataset....")
```

```
End of Practice Dataset....
```

In [ ]: